```
In [79]:
import statsmodels.api as sm
from statsmodels.formula.api import ols


# 1. Load your data
#dataset = pd.read_csv('PrePlacement.csv')

# 2. Fit the Two-Way ANOVA model with interaction
# 'C()' tells python these are categorical factors
model = ols('salary ~ (degree_p) * C(workex)', data=dataset).fit()
anova_table = sm.stats.anova_lm(model, typ=2) # Typ 2 handles unequal group sizes well

print("--- ANOVA TABLE ---")
print(anova_table)
```

```
--- ANOVA TABLE ---
                          sum_sq     df          F        PR(>F)
C(workex)           3.206513e+11    1.0  17.345019  4.543345e-05
degree_p            7.194807e+11    1.0  38.918935  2.380450e-09
degree_p:C(workex)  5.050880e+10    1.0   2.732177  9.983226e-02
Residual            3.900683e+12  211.0        NaN           NaN
```

```
#H0=There is no significance difference of (degree_p) &(workex)
#H1= There is significance difference  of (degree_p) &(workex)

#because (2.380450e-09 &0.0000454 ; ie :0.00000000228)which extremly satisfy p<0.05  Reject the NULL Hypothesyis and
accept the Alternative Hypothesyis
```

Both work experience (`workex`) and degree percentage (`degree_p`) have a highly statistically significant impact on salary. However, their interaction is not significant.

Here is the line-by-line breakdown of your new ANOVA table:

1. Main Effect of Degree Percentage (`degree_p`)

- p-value (`PR(>F)`): `2.380450e-09` (This is scientific notation for `0.00000000238`)
- Meaning: This is extremely significant (far below 0.05). Your degree percentage is a powerful predictor of salary. Higher percentages strongly correlate with higher salaries in your dataset.

2. Main Effect of Work Experience (`C(workex)`)

- p-value (`PR(>F)`): `4.543345e-05` (This is scientific notation for `0.0000454`)
- Meaning: This is also highly significant (well below 0.05). Having work experience vs. not having work experience creates a confirmed, reliable difference in salary.

3. Interaction Effect (`degree_p:C(workex)`)

- p-value (`PR(>F)`): `9.983226e-02` (This is `0.0998` or roughly 10%)

- Meaning: This is not significant (above 0.05). Because this interaction is not significant, it means that the value or boost a person gets from having a higher degree percentage doesn't drastically change whether they have work experience or not. The two factors act independently to increase salary.
